

Contextual Recognition Network: Combining DDPG and Contextual Affordances for Robotic Safe Grasping

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ABSTRACT

With the advancement of deep learning, numerous research initiatives have emerged focusing on enabling robots to identify and retrieve target objects within complex domestic environments. However, current research lacks effective integration of contextual affordances information in robotic systems. This paper introduces an intelligent grasping system to facilitate object prediction and safe policy learning for home-use robots. Particularly, we introduce the Context Recognition Network (CRN) to predict the potential failure likelihood of each action. We develop a grasping system based on DDPG (Deep Deterministic Policy Gradient) as the benchmark. We compare the benchmark's performance with that of the CRN-equipped grasping system. Our results indicate that the CRN-equipped grasping system outperforms DDPG by blocking failure action and instead choosing an appropriate pose based on the object prediction to retrieve the object with fewer computational resources.

CCS CONCEPTS

• Computer systems organization → Robotic autonomy.

KEYWORDS

complex domestic environments, contextual affordances, vision, deep reinforcement learning

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1 INTRODUCTION

Recently, the development of robotic systems applied in domestic environments has progressed rapidly [6]. Current grasping systems capture contextual information through RGB images [8]. With the contextual information, deep learning algorithms could be used to build grasping systems to address tasks in complex home settings. Two of the main current grasping systems are based on neural networks and reinforcement learning.

Neural network-based systems like a combination of CNN and Transformer, could build a highly accurate grasping system [10]. However, they suffer from high computational costs, especially for real-time scenes [8].

As for reinforcement learning-based systems, we focus on deep reinforcement learning, which has high accuracy by combining reinforcement learning and deep neural networks for approximating the policy with neural networks [2]. However, recognition failures and obstacle crashes are common for reinforcement learning agents, posing risks to humans and furniture in home settings [11]. The adaptability of deep reinforcement learning-based systems is also limited [16]. Contextual affordance could be a potential solution for object prediction for novel objects and ensure training safety by preventing dangerous actions resulting from recognition failure [5].

Contextual affordance [4] describes the potential actions of an object based on its properties like shape. Hence, grasping systems can select the optimal grasping pose from the potential actions for grasping novel objects [12]. Furthermore, contextual affordance can be applied in various scenarios and provide actionable contextual information for the downstream tasks [9, 15].

In this paper, we aim to develop an intelligent robotic arm grasping system for UR5 as shown in Figure 1, it integrates contextual affordance and DDPG for safe grasping without relying on substantial training data. Our contributions are as follows:

- We develop an intelligent grasping system with contextual affordance that allows robotic arms to grasp novel objects.
- We apply a vision system to the deep reinforcement learning network for safe grasping.
- We effectively integrate contextual affordance information into our intelligent robotic arm vision grasping system for strong adaptability to unfamiliar and complex home environments.



Figure 1: Simulation environment of a kitchen desk. In this paper, we simulate a home desk environment where the robotic arm needs to grasp cups with different color features on the desk. From left to right, the first image shows the initial state of the simulated environment. The second image shows the environment after the robotic arm executed an action that led to a failure state, causing damage to the environment. The third image demonstrates how our CRN successfully blocked the robotic arm from executing an action that would lead to failure. The fourth image shows how, after incorporating the CRN, the DDPG re-planned a motion trajectory for the robotic arm that successfully and safely grasped the object, allowing the robotic arm to complete the grasping task successfully.

2 METHOD

2.1 Preliminaries

2.1.1 Context Recognition Network. The Context Recognition Network (CRN) is designed as an Artificial Neural Network (ANN) [1] that inputs the context features and outputs a vector indicating the contextual state of the environment. This state includes the DDPG network’s action failure probabilities, helping to avoid dangerous actions and ensure safe grasping by the robotic arm in complex environments.

The CRN is trained on contextual information to predict failure actions that receive significant negative rewards. Integrating CRN with our DDPG algorithm enhances the robotic system’s awareness of operational context, leading to safer and more robust decision-making. Table 1 shows that the CRN correctly predicted 1748 negative instances and 1680 positive instances. We used 2000 positive samples and 2000 negative samples to train the CRN.

Table 1: Confusion Matrix of CRN

	Predicted Negative	Predicted Positive
Actual Negative	1748	251
Actual Positive	320	1680

Based on the training data, the benchmark DDPG system without CRN avoids dangerous actions 80% of the time. With CRN, the avoidance rate improves to 86.40%.

2.1.2 DDPG Framework. We propose a DDPG network framework designed for the motion trajectory planning of robotic arms [13]. We have adapted the traditional DDPG network framework to accommodate the kinematic principles of robotic arms, our DDPG network structure is specifically tailored for the safe grasping of robotic arms in domestic environments [7].

Algorithm 1 shows the DDPG framework integrated with the Context Recognition Network (CRN). The CRN assesses the probability of failure for each action; if this probability exceeds 50%, the action is considered likely to fail. The CRN is trained with environmental data and failure actions from the training process. It

Algorithm 1 DDPG Algorithm with Context Recognition Network (CRN)

Data: Target object coordinates from the vision system

Result: Optimized motion trajectories for the robotic arm

for each episode do

 Reset environment to get the initial state

 Update target object coordinates from the vision system

 Collect initial environment information for CRN

for each step do

 Select action from Actor with added noise $\mathcal{N}$ (Ornstein-Uhlenbeck, $\theta = 0.15$, $\sigma = 0.2$) based on the coordinates

 CRN evaluates the action for potential failure

if CRN predicts failure **then**

 Plan a new motion trajectory using DDPG and collect failure action

end if

 Execute action in the environment

 Observe new state, reward, and done signal

 Store transition (state, action, reward, new state) in replay buffer (size = 1,000,000)

 Sample a minibatch from replay buffer (size = 64)

 Predict next state and reward using Actor_target and Critic_target

 Update Critic by minimizing the loss:

$$L = \mathbb{E} [(r + \gamma Q'(s', \pi'(s')) - Q(s, a))^2] \text{ where } \gamma = 0.99$$

 Update Actor using the sampled policy gradient:

$$\nabla_{\theta\pi} J = \mathbb{E} [\nabla_a Q(s, a | \theta^Q) |_{a=\pi(s|\theta^\pi)} \nabla_{\theta\pi} \pi(s|\theta^\pi)]$$

 Perform soft update of the target networks:

$$\theta' \leftarrow \tau\theta + (1 - \tau)\theta' \text{ where } \tau = 1 \times 10^{-3}$$

end for

end for

receives actions predicted by the DDPG framework and estimates their failure probabilities. This integration prevents the execution of unsafe actions in home settings, thereby enhancing safety and accelerating the training progression [3].

The Mean Squared Bellman Error (MSBE) is utilized as the loss function, which effectively quantifies the difference between the current estimated Q-values and the target Q-values obtained from the target networks.

$$L(\theta^Q) = \frac{1}{N} \sum_{i=1}^N \left(y_i - Q(s_t, a_i | \theta^Q) \right)^2, \quad (1)$$

where y_i is the target Q-value computed using the reward received after taking action a_i in the state s_t , and N is the number of samples in the minibatch, as defined in Equation (1).

To enhance exploration, particularly in the complex and unpredictable environment of a home, Ornstein-Uhlenbeck noise is added to the action selection process:

$$x_{t+1} = x_t + \theta(\mu - x_t)\Delta t + \sigma\sqrt{\Delta t}\mathcal{N}(0, 1), \quad (2)$$

where $\theta = 0.15$ is the rate of mean reversion, $\mu = 0$ is the mean, $\sigma = 0.2$ is the volatility, $\mathcal{N}(0, 1)$ is Gaussian noise with mean 0 and variance 1, and Δt is the time step, set to 1 in our implementation, as defined in Equation (2). This noise model generates temporally correlated exploration actions, which helps the agent explore the action space more effectively.

2.1.3 Reward Function. In our proposed robotic grasping system, the reward function is designed to leverage contextual affordances to enhance safety and efficiency. The combined reward function r_t at time step t incorporates both distance-based incentives and contextual penalties to guide the robot's behavior:

$$\begin{aligned} r_t &= r_{\text{distance}} + r_{\text{context}}, \\ r_{\text{distance}} &= -\alpha \|\text{goal_pose}[:3] - \text{current_pose}[:3]\|, \\ r_{\text{context}} &= \begin{cases} -C & \text{if the action leads to a failure state} \\ 0 & \text{otherwise} \end{cases}, \end{aligned} \quad (3)$$

where r_{distance} penalizes distance from the target, encouraging the robotic arm to move closer to the desired goal position. r_{context} applies a penalty [4] when the robot's actions could lead to failure states. Both components are integral to the combined reward function represented in Equation (3), which balances the objectives of reaching the target efficiently while avoiding unsafe actions.

Contextual affordances are also integrated into the state representation to guide safe actions effectively. The state s_t includes joint angles, end-effector pose, and contextual features that collectively inform the decision-making process of the robotic system, ensuring both efficient and safe operations.

2.2 Proposed Approach

Our proposed approach leverages a vision-based system for robotic grasping powered by deep reinforcement learning. Our architecture, illustrated in Figure 2, includes two systems designed to enhance the robot's interaction with its environment through precise object detection and grasp synthesis.

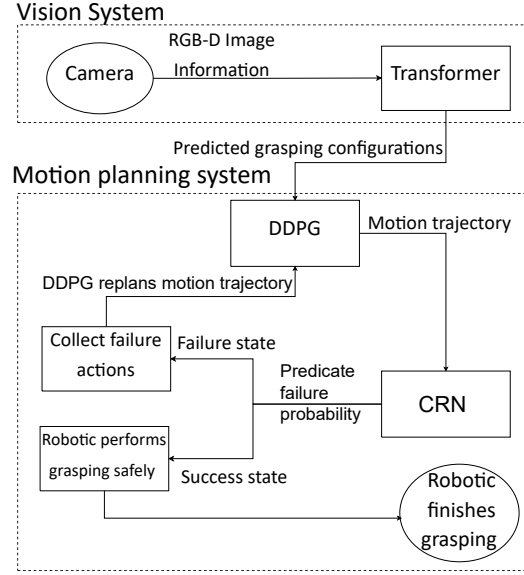
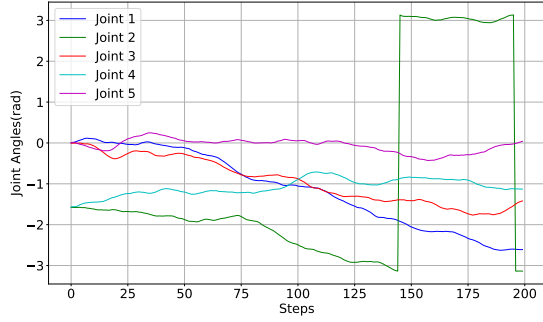


Figure 2: Proposed architecture. We propose a vision-based grasping system underpinned by a Deep Deterministic Policy Gradient (DDPG) network designed for grasping tasks. Our system utilizes an RGB-D camera to capture the image information of the target object. This image information is then processed by a deep-learning visual network that incorporates a transformer architecture, enabling the prediction of grasping configurations. Subsequently, the predicted grasping configuration is sent to the DDPG network for further motion trajectory planning.

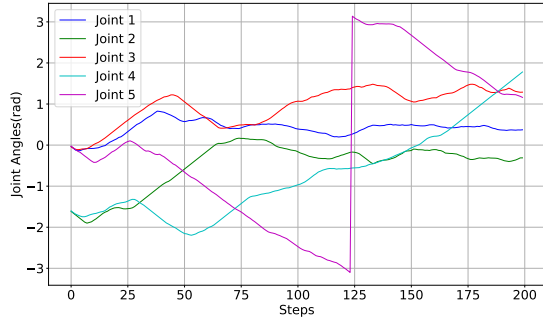
The initial system is the Vision System, where we employ an "eye-in-hand" [14] RGB-D camera to capture the raw image information of the object to grasp. The transformer processes the raw image and outputs the optimal grasping points and poses, which will be passed to the Motion Planning System. The Motion Planning System starts with the DDPG network, which uses the grasping configuration from the Vision System to plan and pass the potential grasping trajectories to CRN. CRN will decide whether the potential grasping trajectories are in a failure state or a success state. In a failure state, the current trajectories will be rejected and new trajectories will be created by DDPG for assessment. Meanwhile, the parameters of this current failure state will be stored and used to train the CRN model. If it is a success state, then the robot will finish the grasping with the approved trajectories.

3 RESULT

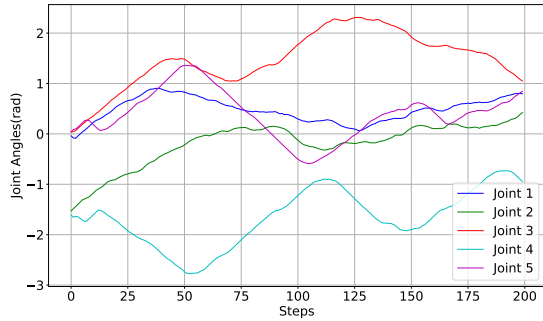
We document the angle of each robotic arm joint at every step. Figure 3a illustrates the motion trajectories planned by the DDPG network for each joint of the robotic arm at the initial stage of training. Specifically, Figure 3b shows the trajectories planned without using the CRN in the final step of training, while Figure 3c depicts the trajectories planned by the network with CRN in the final iteration of training.



(a)



(b)



(c)

Figure 3: Robotic arm joint data (a) First step. (b) Final step without CRN. (c) Final step with CRN. Figure 3a shows the robotic arm joint data collected at the initial stage of training, while Figure 3b and Figure 3c show the robotic arm joint data collected at the final step without CRN and with CRN, respectively.

In Figure 3a, we can see the pink line, which is joint5, has significant fluctuations in the initial training phase. When the robotic arm performs dangerous actions, it often damages the surrounding environment and causes significant twists in its joints. Therefore, we observe substantial changes in the robotic arm joint data during failed actions. Although the CRN network is used to identify failed

actions, it has not yet fully adapted to the collected contextual information data in the early stages of training. In Figure 3b, we can see that by the end of the training phase, the angles of the robotic arm's joints still exhibit significant fluctuations, this is because without the use of CRN, although we have finished all the training iterations, the DPPG is still hard to plan a safe trajectory for the robotic arm. From Figure 3c, we can observe that there are no big fluctuations because the CRN has undergone multiple rounds of training and can better predict failed actions, thereby preventing the robotic arm from causing a failure state and damaging the environment, resulting in significant changes in joint angles.

After completing all training cycles, the simulation result is shown in Figure 4. This plot shows the experimental results of 10 agents trained using DDPG with CRN and DDPG without CRN in a contextual environment. The red curve represents the agent incorporating CRN, the blue curve represents the agent without CRN, and the shadowed area around the curves shows the standard deviation for ten agents. We compute the total collected reward R_T , defined as the sum of all individual rewards r_i received during the learning process, i.e., $R_T = \sum_{i=1}^N r_i$ with $N = 700$. The accumulated R_T values for the DDPG method with and without CRN are 4641 and -3984, respectively. It can be observed that the DDPG algorithm incorporating CRN significantly enhanced the convergence speed of the reinforcement learning and also stabilized performance in the later stages of training. In contrast, the DDPG algorithm without contextual affordances showed no signs of convergence up to 700 steps. This is due to the lack of contextual information to identify and avoid actions that lead to failures.

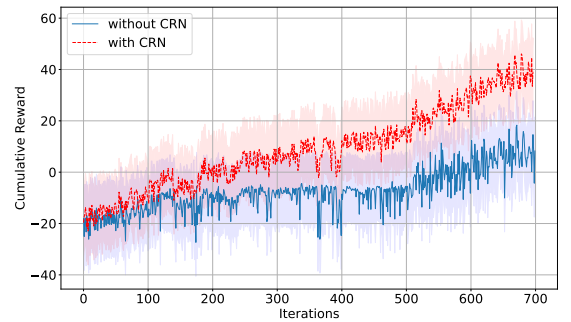


Figure 4: Ten agents cumulative reward. The dashed red curve and the solid blue curve depict the mean cumulative rewards obtained by agents with and without the CRN, respectively.

4 CONCLUSION

The research completed in this paper has successfully integrated contextual affordances into reinforcement learning algorithms. Furthermore, the effectiveness of the architecture proposed in this work has been validated in a simulated environment. Overall, the CRN network significantly accelerates the convergence speed of reinforcement learning and facilitates safe grasping by the robotic arm in a domestic environment. In the future, we will transfer our method to a real-world UR5 robotic arm, and address the sim2real problem.

REFERENCES

- [1] S Agatonovic-Kustrin and Rosemary Beresford. 2000. Basic concepts of artificial neural network (ANN) modeling and its application in pharmaceutical research. *Journal of pharmaceutical and biomedical analysis* 22, 5 (2000), 717–727.
- [2] Chengjun Chen, Hao Zhang, Yong Pan, and Dongnian Li. 2024. Robot Autonomous Grasping and Assembly Skill Learning Based on Deep Reinforcement Learning. *The International Journal of Advanced Manufacturing Technology* 130, 11 (2024), 5233–5249.
- [3] Francisco Cruz, Sven Magg, Cornelius Weber, and Stefan Wermter. 2014. Improving Reinforcement Learning with Interactive Feedback and Affordances. In *4th International Conference on Development and Learning and on Epigenetic Robotics*. IEEE, IEEE, Palazzo Ducale, Genoa, Italy, 165–170.
- [4] Francisco Cruz, Sven Magg, Cornelius Weber, and Stefan Wermter. 2016. Training Agents with Interactive Reinforcement Learning and Contextual Affordances. *IEEE Transactions on Cognitive and Developmental Systems* 8, 4 (2016), 271–284.
- [5] Francisco Cruz, German Ignacio Parisi, and Stefan Wermter. 2016. Learning contextual affordances with an associative neural architecture. In *Proceedings of the European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*. Deakin University, Bruges, Belgium, 665–670.
- [6] Marwan Qaid Mohammed, Kwek Lee Chung, and Chua Shing Chyi. 2020. Review of Deep Reinforcement Learning-Based Object Grasping: Techniques, Open Challenges, and Recommendations. *IEEE Access* 8 (2020), 178450–178481.
- [7] Jyotindra Narayan, Shivansh Mishra, Gaurav Jaiswal, and Santosha K. Dwivedy. 2020. Novel Design and Kinematic Analysis of a 5-DOFs Robotic Arm with Three-Fingered Gripper for Physical Therapy. *Materials Today: Proceedings* 28 (2020), 2121–2132.
- [8] Rhys Newbury, Morris Gu, Lachlan Chumbley, Arsalan Mousavian, Clemens Eppner, Jürgen Leitner, Jeannette Bohg, et al. 2023. Deep learning approaches to grasp synthesis: A review. *IEEE Transactions on Robotics* 39, 5 (2023), 3994–4015.
- [9] Chuanruo Ning, Ruihai Wu, Haoran Lu, Kaichun Mo, and Hao Dong. 2023. Where2Explore: Few-shot Affordance Learning for Unseen Novel Categories of Articulated Objects. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., New Orleans, USA, 4585–4596. https://proceedings.neurips.cc/paper_files/paper/2023/file/0e7e2af2e5ba822c9ad35a37b31b5dd4-Paper-Conference.pdf
- [10] Jinxing Niu, Shuo Liu, Hanbing Li, Tao Zhang, and Lijun Wang. 2023. Grasp Detection Combining Self-Attention with CNN in Complex Scenes. *Applied Sciences* 13, 17 (2023), 9655.
- [11] Alex Ray, Joshua Achiam, and Dario Amodei. 2019. Benchmarking Safe Exploration in Deep Reinforcement Learning. *arXiv preprint arXiv:1910.01708* 7, 1 (2019), 2.
- [12] Azhar Aulia Saputra, Chin Wei Hong, and Naoyuki Kubota. 2019. Real-time Grasp Affordance Detection of Unknown Object for Robot-Human Interaction. In *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*. IEEE, Bari, Italy, 3093–3098.
- [13] Hiba Sekkat, Smail Tigani, Rachid Saadane, and Abdellah Chehri. 2021. Vision-based robotic arm control algorithm using deep reinforcement learning for autonomous objects grasping. *Applied Sciences* 11, 17 (2021), 7917.
- [14] Eugene Valassakis, Kamil Dreczkowski, and Edward Johns. 2022. Learning Eye-in-Hand Camera Calibration from a Single Image. In *Conference on Robot Learning*. PMLR, London, UK, 1336–1346.
- [15] Ruihai Wu, Yan Zhao, Kaichun Mo, Zizheng Guo, Yian Wang, Tianhao Wu, Qingnan Fan, Xuelin Chen, Leonidas Guibas, and Hao Dong. 2022. VAT-Mart: Learning Visual Action Trajectory Proposals for Manipulating 3D Articulated Objects. *arXiv:2106.14440 [cs.CV]*
- [16] Zhuangdi Zhu, Kaixiang Lin, Anil K. Jain, and Jiayu Zhou. 2023. Transfer Learning in Deep Reinforcement Learning: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 11 (2023), 13344–13362. <https://doi.org/10.1109/TPAMI.2023.3292075>